Journal of Hazardous Materials 369 (2019) 465–473

Contents lists available at ScienceDirect

Journal of Hazardous Materials

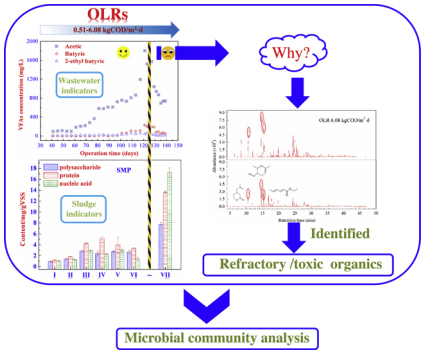
journal homepage: www.elsevier.com/locate/jhazmat

Performance of microaeration hydrolytic acidification process in the pretreatment of 2-butenal manufacture wastewater

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GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
2-Butenal manufacture wastewater
Microaeration hydrolytic acidification process
Organic loading rates
Extracellular polymeric substance
Microbial community

ABSTRACT

The performance of the microaeration hydrolytic acidification (MAHA) process and microbial community were investigated under different organic loading rates (OLRs) for the pretreatment of 2-butenal manufacture wastewater (2-BMW). Results indicated that OLRs had different impact on the performance of MAHA process. More than 23.7 ± 2.3% of the chemical oxygen demand (COD) removal and the highest acidification degree (20.9 ± 3.1%) were obtained when OLRs were less than 4.0 ± 0.1 kgCOD/m^3 d. However, further increasing OLRs to 6.1 ± 0.1 kgCOD/m^3 d subsequently led to the significant reductions of COD removal and acidification degree. In addition, it could be preliminarily inferred that 2H-pyran-2-one tetrahydro-4-(2-methyl-1-propen-3-yl), 5-formyl-6-methyl-4,5-dihydropyran and ethyl sorbate were the main refractory and toxic organics for microorganisms in the wastewater. The soluble microbial product (SMP) and extracellular polymeric substance (EPS) contents (protein, polysaccharide, nucleic acid) had obvious changes under different OLRs. With parallel factor (PARAFAC) model, four fluorescent components were identified. The F_{max} of protein-like substances in SMP significantly decreased with increasing OLRs to 6.1 ± 0.1 kgCOD/m^3 d, which might attribute to fluorescence quenching. Illumina MiSeq sequencing revealed that *Pseudomonas*, *Longilinea*, T78, *Clostridium*, WCHB1-05, *Acinetobacter*, SHD-231 and *Oscillospira* were dominant genera at different OLRs.

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https://doi.org/10.1016/j.jhazmat.2019.02.034
Received 17 October 2018; Received in revised form 27 January 2019; Accepted 9 February 2019
Available online 15 February 2019
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1. Introduction

2 -Butenal (also named crotonaldehyde) is an important industrial chemical for synthesizing sorbic acid. Sorbic acid as food preservative, which is recognized as safe substance in certain degree, has become one of the most common preservatives in the food industry throughout the world [1]. With increasing demand on sorbic acid, the yield of 2-butenal presents an increasing trend. In China, 2-butenal is mainly produced by aldehyde condensation technology [2]. With this production technology, aldehyde, acetic acid and polymerization inhibitor are needed during the production process, which leads to a toxic, refractory and highly concentrated 2-butenal manufacture wastewater (2-BMW). Untreated wastewater released directly into the biological treatment processes could cause severe damage to microbial activity. At present, the effective technologies for wastewater pretreatment have been rarely reported in the literature. Thus, it is necessary to explore an efficient and economical pretreatment process for 2-BMW.

Hydrolytic acidification (HA) is an economical and effective pretreatment way, which plays an important role in breaking refractory/toxic organic matters and improving the biodegradability [3,4]. Therefore, HA process has been widely used in pretreatment field of toxic and refractory wastewater [5]. However, hydrolysis is usually a rate-limiting step for wastewater containing complex pollutants. To solve this situation, some new methods for enhancing hydrolysis efficiencies were proposed. Recently, literature reported that microaeration was an alternative method to enhance hydrolysis of complex organic matter [6]. Under anaerobic conditions metabolites (lactic acid and ethanol) are usually formed to oxidise nicotinamide adenine dinucleotide ($NADH_2$). However, microaeration could effectively reduce the formation of toxic metabolites and increase formation of the desired products [7]. In addition, microaeration could help in the synthesis of certain lipids that are required for the cellular growth [8]. Moreover, the study of Charles et al. revealed that microaeration could enhance the cellulase and protease exoenzyme activities [9]. Until now, microaeration has been predominantly used for the desulphurization of biogas [10]. Besides, some researches have examined the effect of microaeration on the solid waste anaerobic digestion [11,12]. However, there are few research available with regard to the application of microaeration hydrolytic acidification (MAHA) process on the pretreatment of industrial wastewater.

A lot of researches have been carried out on the effect of different factors on anaerobic process rather than HA process. Organic loading rate (OLR) is commonly reported to be a critical parameter in anaerobic process. However, for 2-BWM, the effect of OLRs on the pretreatment efficiency remains unknown with MAHA process. Obviously, the stability of reactor performance is susceptible to toxic shock from the pollutants in 2-BWM. The toxic shock to microorganisms became greater with increasing OLRs. Therefore, it is necessary to investigate the effect of OLRs on the reactor performance for MAHA process. In addition, it is reported that the production of microbial extracellular polymeric substance (EPS) is considered to be an important self-protection strategy under unfavorable environment [13]. For example, Shi et al. reported that granules produced more EPS with adding tetracycline (TC), and TC caused EPS fluorescence spectroscopy quenching [14]. Nevertheless, the effect of industrial wastewater on the change of EPS content under different OLRs has been relatively less studied, especially for the 2-BWM. The microbial community is crucial to improve efficiency of biological treatment process. Several studies have investigated the microbial community during HA process treating dyeing wastewater [15], 2-BMW [16], and trimethylolpropane [17]. However, there is no related literature reporting the microbial community in MAHA process for 2-BMW.

The main objective of this work was to evaluate the effect of OLRs on the bioreactors performance by HAHA process. Firstly, the effects of OLRs on the bioreactor performance were investigated. Secondly, variations of soluble microbial product (SMP) and the EPS contents and fluorescence characteristics of EPS under different OLRs were analyzed. Thirdly, microbial community structures with OLRs were investigated. This study would provide reference for practical application of MAHA process for the pretreatment of 2-BMW.

2. Material and methods

2.1. Main characteristics of 2-BMW

The wastewater used in this work was collected from the production processes of 2-butenal in northern China. The physicochemical characteristics of 2-BMW are as following: pH 3.8 ~ 4.3, chemical oxygen demand (COD) 86,702 ~ 87,945 mg/L, biochemical oxygen demand over five days (BOD_5) 10,907 ~ 11,614 mg/L (BOD_5 value was measured after wastewater dilution), total organic carbon (TOC) 29,101 ~ 29,315 mg/L, alkalinity 230 ~ 257 mg/L as $CaCO_3$, total dissolved solids (TDS) 15,901 ~ 16,285 mg/L.

2.2. Experimental setup and inoculum

The expand granular sludge bed (EGSB) reactor was made of acrylic plexiglass with an internal dimeter of 60 mm and height of approximately 1600 mm. The reactor was composed of reaction ozone (4.5 L) and sedimentation ozone (4.8 L) (Fig. S1). The temperature was maintained at 37 ± 1 °C by a thermostatic waterbath. The hydraulic retention times (HRT) was 24 h. The up-flow velocity (V_{up}) was kept stable at 2.9 m/h by effluent recirculation with peristaltic pump. Air was added continuously in the effluent tank by aerator pump, which was carried into the EGSB reactor through effluent recirculation. The microaeration of 0.2 ~ 0.5 mg/L was provided to the EGSB reactor. The EGSB reactor was inoculated with 2.0 L anaerobic sludge, which was obtained from the integrated petrochemical wastewater treatment plant. The concentration of suspended solid (SS) and volatile suspended solid (VSS) were 11.1 ± 0.5 g/L and 8.0 ± 0.3 g/L, respectively. The volatile suspended solids to suspended solids (VSS/SS) of sludge was 0.72.

The influent COD was gradually increased by diluting 2-BMW. NH_4Cl and KH_2PO_4 were added in the wastewater as nitrogen and phosphorus, respectively. The nutrient ratio (COD:N:P) used was 200:5:1.

2.3. Analytical methods

Analysis of COD, TDS, alkalinity and VSS were analyzed according to the standard methods for the examination of water and wastewater [18]. TOC-V_{CPH} instrument (Shimadzu, Japan) was employed to measure TOC. The pH and oxidation reduction potential (ORP) were determined by a portable meter (FE20, Mettler Toledo, Switzerland). Volatile fatty acids (VFAs) and identification of organic matters were determined in accordance with previous reports [16].

2.4. EPS extraction and analysis

A heat extraction method was employed to extract SMP, loosely bound-EPS (LB-EPS) and tightly bound-EPS (TB-EPS) from sludge samples under different OLRs [19].

The polysaccharide (PS) content in SMP, LB-EPS and TB-EPS extractions was measured with phenol-sulfuric acid method [20]. The protein (PN) content was determined using BCA protein kit (Sangon, China). The nucleic acid (NA) content was measured using the diphenylamine colorimetric method with sodium salt of calf thymus DNA as the standard.

2.5. 3D-EEM fluorescence spectroscopy and PARAFAC analysis

Three-dimensional excitation-emission matrix (3D-EEM)

fluorescence spectra of SMP, LB-EPS and TB-EPS extractions were obtained using a luminescence spectrometry (F-7000, Hitachi, Japan). The excitation wavelength (Ex) was set as 200 ~ 450 nm with emission wavelength (Em) 280 ~ 550 nm.

Due to the complexity of SMP and EPS compositions, there was the problem of significantly overlapped fluorescence spectra. Therefore, parallel factor (PARAFAC) analysis was adopted to separate the actual fluorescence spectra from the 3D-EEM [21,22]. The spectrum of double distilled water as blank sample was measured, and was deducted from the 3D-EEM spectra of the samples. To eliminate the distractions of Rayleigh scattering, the 3D-EEM data close to the Rayleigh scattering line were set to zero [23]. The corrected fluorescence data were then modeled with the DOMFluor Toolbox in Matlab R2009a [24]. The split-half and residual analysis was further performed to validate the number of fluorescence components [25]. The fluorescence intensity of each component was represented by F_{max} (R.U., i.e. Raman units) [26].

2.6. Microbial structure analysis

In this study, sludge samples were taken in the stable operation phase with different OLRs. DNA was extracted from sludge samples with Tiangen® soil DNA kit (Tiangen Biotech, China) following the manufacture's protocols. The extracts were stored at −20 °C before further analysis. The V3-V4 regions of the 16 s rDNA genes were amplified with 340 F (CCTACGGGNBGCASCAG) and 805R (GACTACNVGGGTATCTAATCC). All polymerase chain reaction (PCR) reaction mixture was purified with the MinElute Gel Extraction Kit (Qiagen, Germany). The Invitrogen™-Qubit fluorometer quantitative system (California, USA) was used to quantity the amplicons. Then high-throughput sequencing of 16S DNA amplicons were sequenced with the Illumina MiSeq platform at SinoGenoMax Co., Ltd (Beijing, China).

3. Results and discussion

3.1. Performances of EGSB reactor under different OLRs

The reactor was ran continuously for 159 days and experienced shock loading and recovery stage. Fig. 1 shows the reactor performance in terms of COD, COD removal, pH, ORP, VFAs concentration and acidification degree during the experiment.

As shown in Fig. 1 (a), OLRs increased gradually from 0.5 ± 0.1 to 6.1 ± 0.1 kgCOD/m^3 d with influent COD increasing from 512 ± 25 to 6037 ± 111 mg/L at shock loading stage. To reactivate microbial activity, OLR was then decreased to 4.1 ± 0.1 kgCOD/m^3 d corresponding to influent COD 4082 ± 94 mg/L at recovery stage. The effect of different OLRs on the COD removal is displayed in Fig. 1(b). The COD removal was 61.7 ± 5.6% at stage I. From stage II to VI, the COD removal decreased steadily from 44.1 ± 4.5% to 23.7 ± 4.5% with OLRs increasing from 1.1 ± 0.1 to 4.0 ± 0.1 kgCOD/m^3·d. However, further increasing OLRs to 6.1 ± 0.1 kgCOD/m^3 d (stage VII), the COD removal decreased sharply to 13.4 ± 1.8%. At recovery stage VIII, OLR was decreased to 4.1 ± 0.1 kgCOD/m^3·d by reducing influent COD. The COD removal quickly achieved 23.1 ± 1.1%. Comparatively, the COD removal in the HA reactor remained less than 20% when OLR was decreased to 4.1 kgCOD/m^3 d after shock loading of 6.1 kgCOD/m^3 d [16].

The suitable pH and ORP values could provide good growing conditions for microorganisms, so the above two parameters were monitored in the course of the experiment [27]. The Fig. 1(c) depicts the variations of pH and ORP under different OLRs. During stage I ~ II, the effluent pH maintained 5.8 ~ 6.7 while the reactor interior ORP rapidly lowered from −55 mV to −315 mV. The reactor created relatively good circumstance for microorganisms. As increasing OLR (stage III), the effluent pH decreased gradually from 6.1 to 5.1. This implies that the reactor was not well self-buttered. Meanwhile, ORP increased steeply from −322 mV to −239 mV. To resolve the above problems, $NaHCO_3$ as the alkalinity was added to influent for the following experiment. After improving the influent alkalinity, the effluent pH soon returned to normal and maintained in the range of 5.9 ~ 7.4 at stage IV ~ VI. In addition, ORP lowed quickly to below −300 mV. Despite the slight fluctuation of effluent pH and reactor interior ORP, the overall performance of EGSB reactor was steady. However, further improving OLRs, reactor interior ORP increased dramatically from −269 mV to −79 mV, possibly due to the increase of inhibition and/or toxicity effects on microorganisms. Similar result has been found previously by Blanc et al., who indicated that an increasing ORP correlated to inhibited or decreasing levels of digester performance [28]. At recovery stage VIII, reactor interior ORP rapidly dropped to −340 mV with OLR decreasing to 4.1 ± 0.1 kgCOD/m^3 d.

Fig. 1(d) shows the VFAs concentration and acidification degree under different OLRs. Seven kinds of VFAs, including acetic, propionic, *n*-butyric, *n*-valeric, *n*-hexanoic, 2-methyl butyric and 2-ethyl butyric were detected in this study. However, the dominant fermentation products were acetic, *n*-butyric, and 2-ethyl butyric. The others VFAs were produced in insignificant amounts. As shown in Fig. 1(d), no sign of VFAs in the effluent was detected at stage I ~ II, which demonstrated that fermentation products were metabolized by facultative bacteria in the reactor. As can be seen in this study, as OLRs increased, the concentration of fermentation products appeared firstly increased and then decreased during stage III ~ VII. With OLRs increasing from 1.6 ± 0.1 (stage III) to 4.0 ± 0.1 kgCOD/m^3 d (stage VI), the acidification degree increased from −12.7 ± 1.0% to 20.9 ± 4.4%. However, further increasing OLR to 6.1 ± 0.1 kgCOD/m^3 d (stage VII) subsequently led to the significant reductions of acidification degree. The acidification degree reduced from 15.6% to −1.5%, with the corresponding COD removal dropping from 17.9% to 11.7%. The phenomenon indicated that OLR of 6.1 ± 0.1 kgCOD/m^3 d caused the EGSB performance to worsen. Ding et al. also reported that acidification degree began to drop when OLR reached maximum, which was largely ascribed to insufficient of extracellular hydrolytic enzymes [29].

3.2. Characterization of organic compositions

The organic compositions in the influent and effluent under different stages were analyzed by GC–MS. Fig. 2 depicts a representative GC–MS chromatogram for influent and effluent at stage VI. As shown in Fig. 2, the species and abundance of organic matters decreased considerably by MAHA process. Twenty-three kinds of typical compounds were detected in the influent, including aldehydes, alcohols, esters, organic acids and phenols. The relative retention times of the first three highest peak were 10.769, 14.772 and 15.507 min, corresponding to 2H-pyran-2-one tetrahydro-4-(2-methyl-1-propen-3-yl), 5-formyl-6-methyl-4,5-dihydropyran and ethyl sorbate, respectively.

Some new peaks (a ~ q) were displayed in Fig. 2 (b), and these new organic matters could be correspond to hydrolytic intermediates and microorganism metabolites, such as acetic acid, 2-pentanone, 2-methyl-propanoic acid, butanoic acid, 3-methyl-butanoic acid, 2-methyl-butanoic acid, 2-ethyl-butanoic acid, hexanoic acid, 3-methyl-2,4-pentandiol,

3-methyl-3-hexen-2-ol, (E)-1-methoxy-4-hexene, 3-cis-hexenyl methyl carbonate, cytosine, cis-3-hexenyl lactate, 2,4-dihydroxypyridine, cis-2-hexenyl acetate, (E)-2-hexenoic acid. The removal efficiencies of organic compounds at different stages are reported in Table S1. Most of the organic matters could be hydrolyzed into more readily degradable organic matters during stage I~VI, which illuminates that the MAHA process was capable of decomposing refractory and toxic organic matters in 2-BMW. Nevertheless, most of compounds were not effectively degraded at stage VII, including especially 2H-pyran-2-one tetrahydro-4-(2-methyl-1-propen-3-yl), 5-formyl-6-methyl-4,5-dihydropyran and ethyl sorbate. The removal efficiencies of the above three compounds were 65.2 ± 4.6%, 70.0 ± 6.1% and 68.5 ± 2.5%, respectively. Because of a lack of commercial standard, the toxic effects

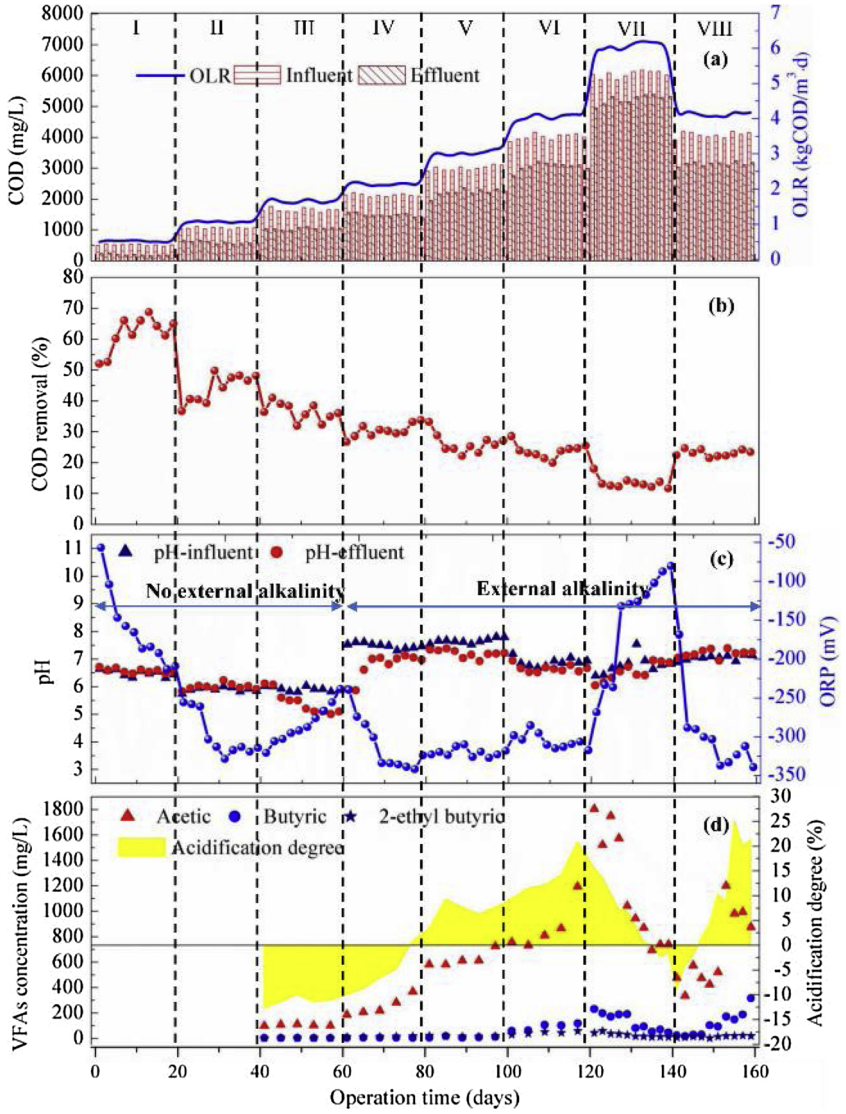


Fig. 1. Changes of influent COD and effluent COD (a), COD removal (b), pH and ORP values (c) and VFAs concentration and acidification degree (d) at different OLRs.

on microorganism of the above these compounds were not measured in this research. Meanwhile, there is no report about the toxicity of 2H-pyran-2-one tetrahydro-4-(2-methyl-1-propen-3-yl) and 5-formyl-6-methyl-4,5-dihydropyran for microorganism. However, the acute toxicity of 2-formyl-3,4-dihydro-2H-pyran has been reported, which has similar structure with 5-formyl-6-methyl-4,5-dihydropyran. The lethal dose (LD_{50}) of 2-formyl-3,4-dihydro-2H-pyran in rat is 4930 mg/kg by oral [30]. In addition, it is reported that sorbate as antimicrobial agent is customarily used to inhibit growth of molds in several food products [31]. Sorbate with the strong hydrophobic property could more easily penetrate the microbial cell membrane, which would disturb microbial physiological function and even poison the microorganism. Therefore, it could be preliminarily inferred that the above three compounds might belong to the refractory and toxic organics by combining molecular structure and literature.

3.3. Variations of SMP and EPS contents under different OLRs

Fig. 3 summarizes the variations of SMP and EPS (LB-EPS, TB-EPS) contents (PN, PS and NA) under different OLRs. The variations of PN, PS and NA in the SMP under different OLRs are shown in Fig. 3(a). With the increase of OLRs, the PN content in the SMP generally increased firstly and then decreased during stage I ~ VI. However, further increase of OLR to 6.1 ± 0.1 kgCOD/m^3 d (stage VII), the PN content increased sharply from 3.4 ± 0.1 to 13.5 ± 0.3 mg/gVSS. The PS and NA contents had no obvious change rule during stage I ~ VI, whereas the PS and NA content increased sharply from 2.7 ± 0.3 to 7.8 ± 0.3 mg/gVSS and 1.4 ± 0.4 to 17.2 ± 0.9 mg/gVSS, respectively, at stage VII. This result might be attributed to release of intracellular material which was caused by toxic substance at stage VII (OLR 6.1 ± 0.1 kgCOD/m^3 d). This phenomenon was similar to the one in literature reported by Fukushi et al., that the PS and NA levels increased in the presence of 40 mg/L of Cu and Cd due to cell death and lysis [32].

The variations of LB-EPS and TB-EPS contents (PN, PS and NA) under different OLRs are shown in Fig. 3(b) and (c), respectively. With the increase of OLRs from 0.5 ± 0.1 to 2.1 ± 0.1 kgCOD/m^3·d, the PN content in LB-EPS and TB-EPS increased from 1.0 ± 0.1 to 6.2 ± 0.9 mg/gVSS and 5.6 ± 0.3 to 21.5 ± 0.8 mg/gVSS, respectively, and the PS content in LB-EPS and TB-EPS increased from 0.5 ± 0.1 to 2.9 ± 0.6 mg/gVSS and 2.5 ± 0.2 to 7.5 ± 0.2 mg/gVSS, respectively. Previous literature demonstrated that the presence of toxic substances stimulated the excretion of EPS production which was an important self-protection strategy of microorganism [33]. Nevertheless, the PN and PS contents in LB-EPS and TB-EPS emerged a

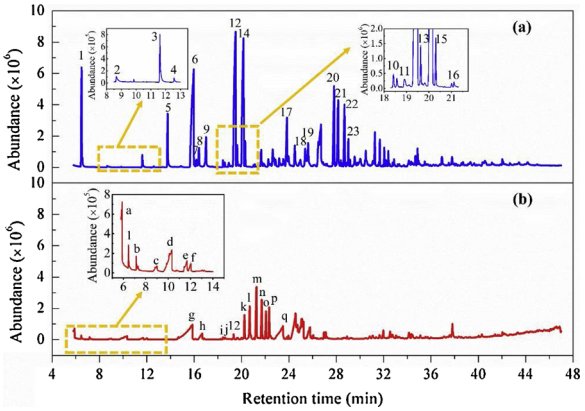


Fig. 2. GC–MS chromatogram of 2-butenal manufacture wastewater samples in influent (a) and effluent (b) at stage VI: (1) 2-butenal; (2) 3-butanolal; (3) 2-ethenyl-2-butenal; (4) 1,3-butanediol; (5) (E,E)-2,4-hexadienal; (6) 2H-pyran-2-one tetrahydro-4-(2-methyl-1-propen-3-yl)-; (7) isopentyloxyethyl acetate; (8) 4-hydroxy-3-methyl-2-butanone; (9) 4-oxo-2-pentenonic acid; (10) 2,4,6-octatrienal; (11) 1-methylbenzyl alcohol; (12) 5-formyl-6-methyl-4,5-dihydropyran; (13) 2-methylbenzaldehyde; (14) ethyl sorbate; (15) spiropentanoic acid, methyl ester; (16) 4,4-dimethylcyclohexadienone; (17) 2-methyl-1-octen-3-yne; (18) methoxyacetic acid, 2-pentyl ester; (19) methoxyacetic acid, 2,2-dimethylpropyl ester; (20) 2-hydroxy-benzeneethanol; (21) 3-n-propylphenol; (22) 3-tert-butylphenol; (23) 1-(2-methoxyphenyl) ethanol; (a) acetic acid; (b) 2-pentanone; (c) 2-methyl-propanoic acid; (d) butanoic acid; (e) 3-methyl-butanoic acid; (f) 2-methyl-butanoic acid; (g) 2-ethyl-butanoic acid; (h) hexanoic acid; (i) 3-methyl-2,4-pentandiol; (j) 3-methyl-3-hexen-2-ol; (k) (E)-1-methoxy-4-hexene; (l) 3-cis-hexenyl methyl carbonate; (m) cytosine; (n) cis-3-hexenyl lactate; (o) 2,4-dihydroxypyridine; (p) cis-2-hexenyl acetate; (q) (E)-2-hexenoic acid.

reducing trend with further increasing OLRs. This results showed that high OLRs might not markedly promote the EPS production due to the increase of toxic substance concentration. Some studies have demonstrated that toxic substance at a high concentration might weaken the bacterial metabolism activity and then reduce the EPS production [34]. It is worth noting that PN content was dominant in EPS production, revealing that PN played a more significant role in protecting cell from the harsh external environment. The observation was also well reported by Sheng et al., in which PN as the major component in EPS protected against the harmful effects by diffusion limitation and/or by chemical binding [35]. As OLRs increased, the NA content in the LB-EPS revealed a general increase, while the NA content in the TB-EPS did not change evidently.

3.4. PARAFAC analysis of 3D-EEM fluorescence spectra

The PARAFAC model of SMP and EPS from sludge samples under different OLRs revealed that the fluorescence could be characterized by four different components (Fig. 4(a)). Fig. 4 (a) also presents Ex/Em loadings of the four components. Component 1 consisted of two excitation maxima at 220 and 280 nm, with one emission peak centered at 355 nm. Component 1 resembled the typical aromatic protein (Ex/Em 220/355 nm) and soluble microbial byproduct-like matter (Ex/Em 280/355 nm) [22]. Component 2 had the maxima loadings at Ex/Em wavelength of 225/305 nm and 265/305 nm, which were associated with tyrosine-like fluorophore. Component 1 and component 2 had been ascribed to protein-like fluorophores [36]. Component 3 had a primary (secondary) fluorescence peak at an Ex/Em wavelength of 275(225)/325 nm, which might be regarded as the blue shift of component 1. Component 4 were characterized by excitation at 210, 240 and 320 nm, and emission at 430 nm, which demonstrated the existence of UVA humic-like substance [37].

PARAFAC analysis also supplied the quantitative information that

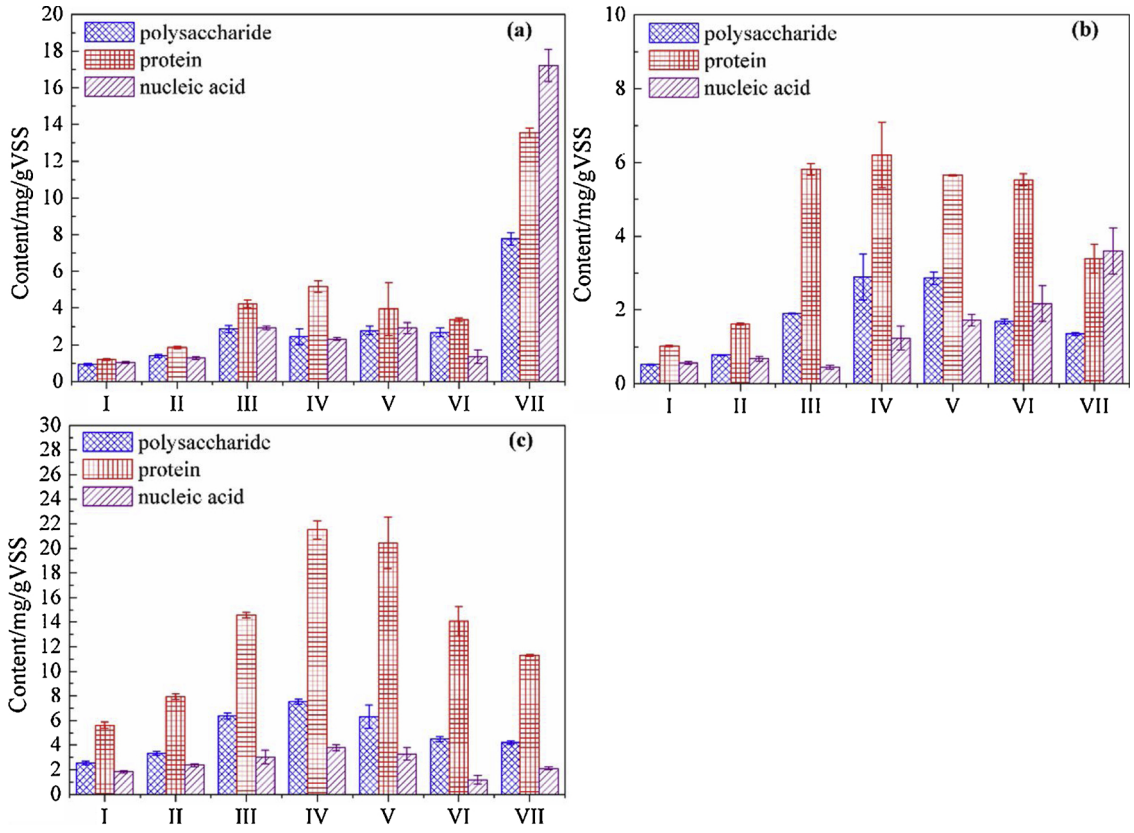


Fig. 3. Variations of SMP content (a), LB-EPS content (b) and TB-EPS content (c) at different OLRs.

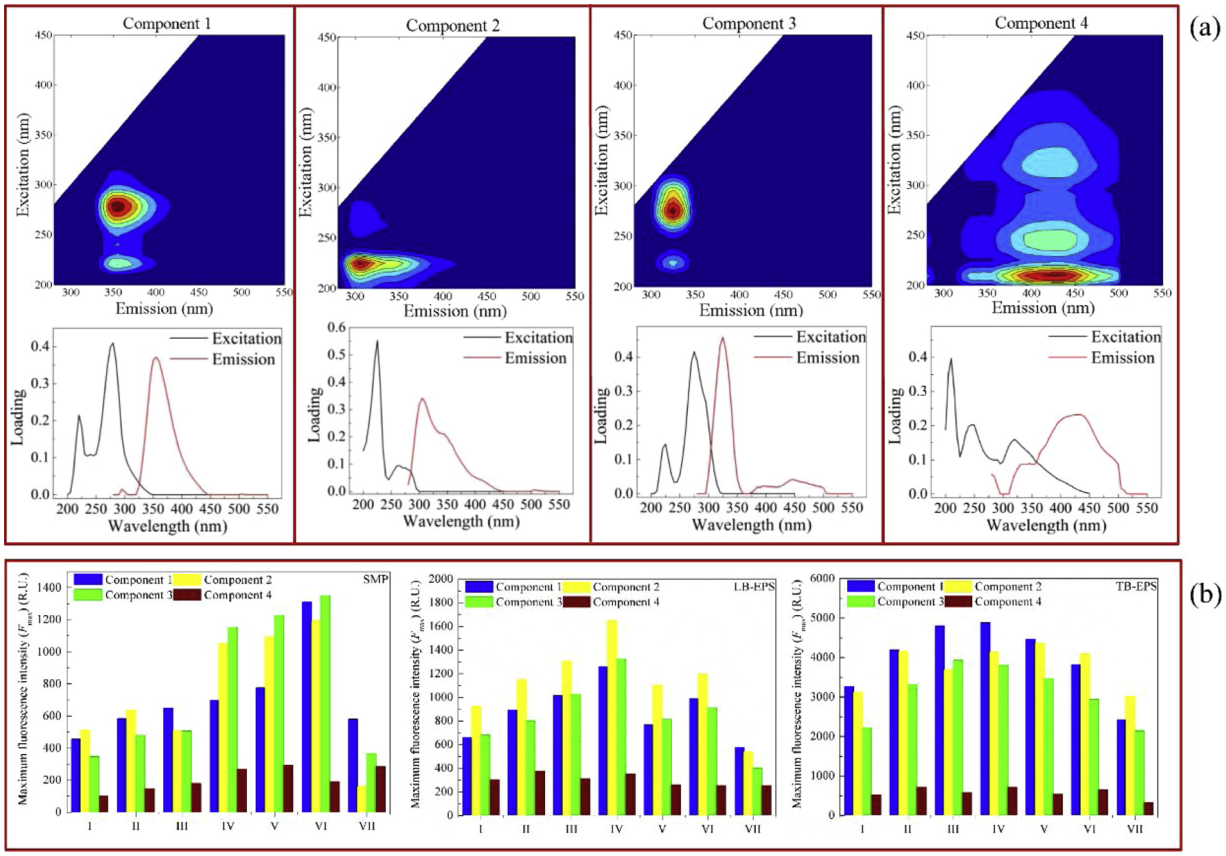


Fig. 4. 3D-EEM spectra of four fluorescence components with PARAFAC analysis (a) and maximum fluorescence intensities (F_{max}) of four PARAFAC-derived components (b).

showed the fluorescence intensity (F_{max}) of the four components in SMP, LB-EPS and TB-EPS samples as OLRs changed. The protein-like substances (component 1, 2 and 3) were major constituents in SMP, LB-EPS and TB-EPS samples. The F_{max} of component 1, 2 and 3 generally showed an increasing trend, which was inconsistent with the protein content in SMP samples during stage I ~ VI. The increase in F_{max} of protein-like substances suggested that it played a certain role in the organic matter removal and transform in biosorption process. Aquino et al. investigated the interaction between protein-like substances and chloroform, indicating that the accumulation of protein-like substances increased as the amounts of toxic chloroform increasing [38]. Nevertheless, under OLR of 6.08 kgCOD/m^3·d (at stage VII), the F_{max} of SMP samples showed a sudden drop, which was the opposite of the protein content change at this stage. Possibly because the protein-like substances of SMP were obviously quenched with organic matters in the wastewater. More detailed, the F_{max} of component 1, 2 and 3 in SMP was decreased from 1310, 1194 and 1351 at stage VI to 580, 161 and 365 at stage VII, respectively. Previous studies have reported that protein-like substances as fluorophores was clearly quenched by phenanthrene [39]. In this present study, the quenching mechanism about fluorescence was not investigated due to intrinsic complexity of wastewater sample. Moreover, comparing to protein-like substances, the F_{max} of humic-like substance (component 4) had no evident change as increasing OLRs, which was probably due to that the contents of humic-like substances were lower than protein-like substances. Similar results have been reported by Hou et al. [40].

The variation tendency of F_{max} of component 1, 2 and 3 in LB-EPS and TB-EPS was different from those in SMP, owing to the fact that LB-EPS and TB-EPS are not susceptible to organic matters as they are less close to 2-BMW than SMP. The F_{max} of component 1, 2 and 3 increased firstly and then decreased in LB-EPS and TB-EPS, which was consistent with the variation of protein. Component 2 was the primary component at stage I ~ VI in LB-EPS, which indicated that there existed a large amount of tyrosine-like substance. The existence of tyrosine-like substance might be a useful emergency measure which was used to avoid the adverse effects of organics in wastewater. For TB-EPS, there was no evident regularity of different components. Su et al. also found that the EEM spectra of the TB-EPS showed no obvious change due to TB-EPS locating in the inner layer of the sludge floc [41].

3.5. Microbial community analysis

Illumina Miseq sequencing was used to analyze the V3-V4 region of 16 s rRNA of sludge samples for revealing the variation of microbial community. The community richness and diversity indices of different samples were shown in Table S2. The OTUs in the samples ranged from 1714 to 4095. The lowest OTUs of 1714 were obtained at stage VII, which exhibited less richness of microbial communities than the one at others stages. Ace and Chao indices indicated community richness, while Shannon and Simpson indices represented community diversity [42]. Ace, Chao and Shannon had significantly changed amount different samples, suggesting that OLRs had an important effect on the microbial community diversity in MAHA reactor. In comparison, lower indices of Ace, Chao and Shannon were achieved at stage VII, which indicated that OLR 6.08 kgCOD/m^3 d (stage VII) had a negative impact on the community richness and evenness.

As shown in Fig. S2, principal component analysis (PCA) revealed obvious separation of microbial communities at different OLRs. The percentage variance of species explained by the first and second axis was 35.59% and 17.99%, respectively. Seven samples were dispersed into four distinct regions of clusters. The sample I and III grouped together, while the samples (II, IV and V) gathered together. They were

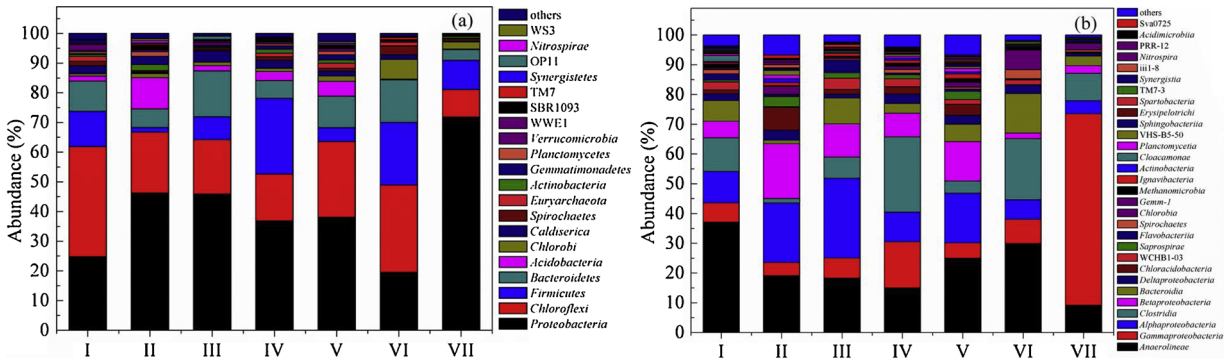


Fig. 5. Relative abundance of the dominant bacterial community at phylum level (a) and class level (b) at different OLRs (relative abundance greater than 0.6% were shown).

well separated from samples (VI and VII), but there was a clear distinction between sample VI and VII. The PCA results further demonstrated that the microbial communities in MAHA reactor were impacted by OLRs.

The diversities of microorganisms at the phylum and class level in MAHA reactors were shown in Fig.5, in which the relative abundance of two taxonomic levels (phylum and class) was greater than 0.6% in at least one sample. As shown in Fig.5 (a), microbial community exhibited similar diversities but obviously different abundance at the phylum level. *Proteobacteria* (19.51 ~ 71.80%), *Chloroflexi* (9.27 ~ 37.06%), *Firmicutes* (1.62 ~ 25.49%) and *Bacteroidetes* (3.59 ~ 15.41%) were dominant phyla in all samples. These four phyla accounted for more than 75% of the total bacteria in seven samples. *Proteobacteria*, *Chloroflexi*, *Firmicutes* and *Bacteroidetes* were reported to be dominant phyla in hydrolytic acidification system for treatment of 2-BMW [16]. Previous research reported that *Proteobacteria* could degrade extensive organic matters and existed widely in wastewater treatment [43]. At stage VII (OLR 6.08 kgCOD/m^3 d), the proportion of *Proteobacteria* reached the maximum (71.80%). In comparison, *Chloroflexi* initially accounted 37.06% at stage I (OLR 0.51 kgCOD/m^3 d) and then decreased to 9.27% at stage VII (OLR 6.08 kgCOD/m^3 d). The study of Lv et al. revealed that *Chloroflexi* could degrade starch to acetate and other short-chain fatty acids [44]. At stage VII, the 2-BMW had an adverse effect on the microbial activity, ultimately resulting in a decrease in the proportion of *Chloroflexi*. The phylum *Firmicutes* not only was a larger participant in VFA generation, but also could secrete more endospores to survive in the toxic environment [45,46]. Meanwhile, *Bacteroidetes* were known to convert amino acids generated during protein hydrolysis into acetate [47]. The relative abundance of *Firmicutes* and *Bacteroidetes* generally appeared increase tendency as OLRs increasing. However, at stage VII they showed a reducing trend, possibly due to the increase of inhibition and/or toxicity effects on microbial activity.

Fig. 5(b) showed the relative abundance of bacterial community on the class level. The dominant bacterial communities were *Anaerolineae* (9.18 ~ 37.06%), *Gammaproteobacteria* (4.47 ~ 64.32%), *Alphaproteobacteria* (4.40 ~ 26.80%), *Clostridia* (1.47 ~ 25.22%), *Betaproteobacteria* (1.90 ~ 18.41%) and *Bacteroidia* (1.23 ~ 13.24%). *Anaerolineae* presented a prominent abundance in all the samples at different stages. Nevertheless, the relative abundance of *Anaerolineae* has been significantly decreased at stage VII. Notably, the relative abundance of *Gammaproteobacteria* was obviously improved, whereas an opposite tendency was shown in the abundance of *Alphaproteobacteria*, *Clostridia* and *Bacteroidia* when OLRs varied from 4.04 to 6.08 kgCOD/m^3·d.

The heatmap graph depicts the profiles of 50 most abundant genera in samples with different OLRs (Fig. 6). There was substantial variation of bacterial community among samples, which implied a great effect of different OLRs on microbial community structure in MAHA process. As shown in Fig. 6, eight major bacterial genera were detected in all samples, namely *Pseudomonas*, *Longilinea*, T78, *Clostridium*, WCHB1-05, *Acinetobacter*, SHD-231 and *Oscillospira*. Under OLRs less than 4.04 kgCOD/m^3·d, the relative abundance of *Pseudomonas* was kept almost stable. However, with OLRs increasing from 4.04 to 6.08 kgCOD/m^3·d, *Pseudomonas* abundance rose from 0.02% to 57.58%. It was reported that *Pseudomonas* was bacterium with the capacity to degrade refractory aromatic compounds [48], and could survive under the circumstance of high-cyanide [49], which suggested that *Pseudomonas* had stronger resistant ability under strict condition. This might be the reason why *Pseudomonas* abundance increased largely with the increase of OLRs. *Longilinea* and *Clostridium* were hydrolysis fermentation and acidogenic bacteria, which played important roles for hydrolysis and acidogenesis during the course of the treatment [50]. The relative abundance of *Longilinea* and *Clostridium* at stage VI accounted for 25.53% and 6.51%, respectively. This result was consistent with the maximum acidification degree at stage VI. Due to higher OLR (stage VII), *Longilinea* and *Clostridium* abundance showed evident decrease, which indicated that the concentration of toxic contaminant at stage VII could be deleterious to *Longilinea* and *Clostridium*. *Oscillospira* with *Firmicutes* was reported to play a crucial role in food waste acidogenic reactors [51]. Moreover, previous study indicated that *Oscillospira* was vulnerable to sulfamethoxazole and its relative abundance decreased with operation time [52]. For T78, there were few reports about its exact role in previous literatures. The *Oscillospira* and T78 abundance overall appeared first increase then decrease trend. In this study, WCHB1-05 and SHD-231 were also detected under different OLRs. According to Petropoulos et al., WCHB1-05 with high lipolysis was often enriched in the absence of available organic material at low temperature [53]. Meanwhile, the genus WCHB1-05 was associated with the degradation of hydrocarbons [54]. SHD-231 was often observed in wastewater, but its function was seldom mentioned. The relative abundance of WCHB1-05 and SHD-231 showed decrease with increasing OLRs in this study. Various studies have reported that *Acinetobacter* had potential ability to degrade toxic substances (3,5,6-trichloro-2-phyidinol, azo dyes, et. al.) [55,56]. The abundance of *Acinetobacter* at stage VII was significantly more than those at other stages, which confirmed that *Acinetobacter* might to be special genera correlated to detoxification.

4. Conclusions

The MAHA process provided an efficient technique for the pretreatment of 2-BMW, but OLR of approximately 6.08 kgCOD/m^3·d would result in a failed performance of the reactor. Additionally, GC–MS analysis results revealed that the refractory and toxic organics in the 2-BMW could be hydrolyzed into more readily degradable organic matters when OLR was less than 4.04 kgCOD/m^3·d. With regard to the microbial community, *Pseudomonas*, *Longilinea*, T78, *Clostridium*, WCHB1-05, *Acinetobacter*, SHD-231 and *Oscillospira* were dominant genera. This study demonstrated that the MAHA process could be

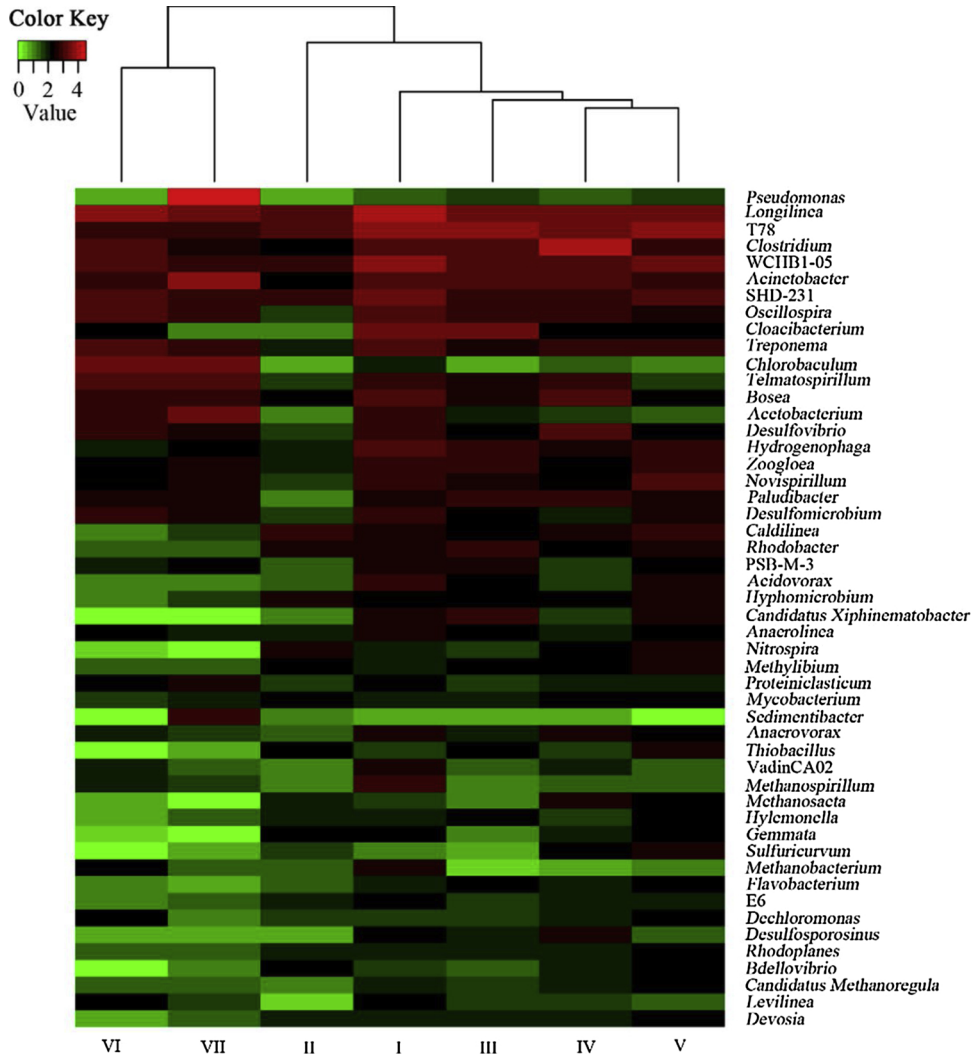


Fig. 6. Heatmap showing 50 most abundant 16S rRNA gene at the genus level with different OLRs (green colors and red colors indicate lower abundance and higher abundance, respectively) (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

employed to pretreatment 2-BMW for reducing shock impact on the following biological treatment.

Acknowledgements

This work was funded by the National Water Pollution Control and Treatment Science and Technology Major Project of China (No. 2017ZX07402002).

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.jhazmat.2019.02.034.

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